System Architecture & Data Flow Diagram

Multi-Tier Pipeline: Data Ingestion -> Vector & Keyword Indexing -> RAG & Multi-Agent -> API & UI

1. Multi-Tier Architecture Pipeline

Tier 1: Data Ingestion & Enrichment	Tier 2: Dual-Index Retrieval Layer	Tier 3: GenAI Intelligence & Multi-Agent
• Yelp Academic Dataset (5.34 GB) • Restaurant Category Filtering • Text Normalization & Cleaning • 8-Category Taxonomy Tagging • Multi-Factor Priority Scoring • SQLite Database (48,288 Records)	• ChromaDB Vector Store (MiniLM-L6-v2) • BM25 Lexical Inverted Index • Reciprocal Rank Fusion (k=60) • Metadata Attribute Filtering • Sub-50ms Candidate Retrieval	• RAG Engine (Gemini 2.5 Flash/Pro) • Agent 1: Classifier & Severity • Agent 2: Root Cause Diagnostics • Agent 3: Actionable SOP Generator • Agent 4: Business Impact Evaluator • Agent 5: LLM Critic & Safety Judge

2. Presentation & Delivery Tier

Tier 4: FastAPI Microservices (RESTful Backend)	Tier 5: Streamlit Operations Dashboard (Frontend)
• Endpoints: /search, /query, /analyze, /prioritize, /agent/workflow, /forecast, /feedback • Pydantic v2 validation & CORS enabled • Automated Swagger API documentation	• 7 Interactive Pages: Executive Dashboard, Search, Analytics, Insights & RAG, Forecasting, Multi-Agent Inspector, Evaluation • Dark mode glassmorphism & Plotly charts